

Engineering Mechanics

Lecture 7: Truss and Frame

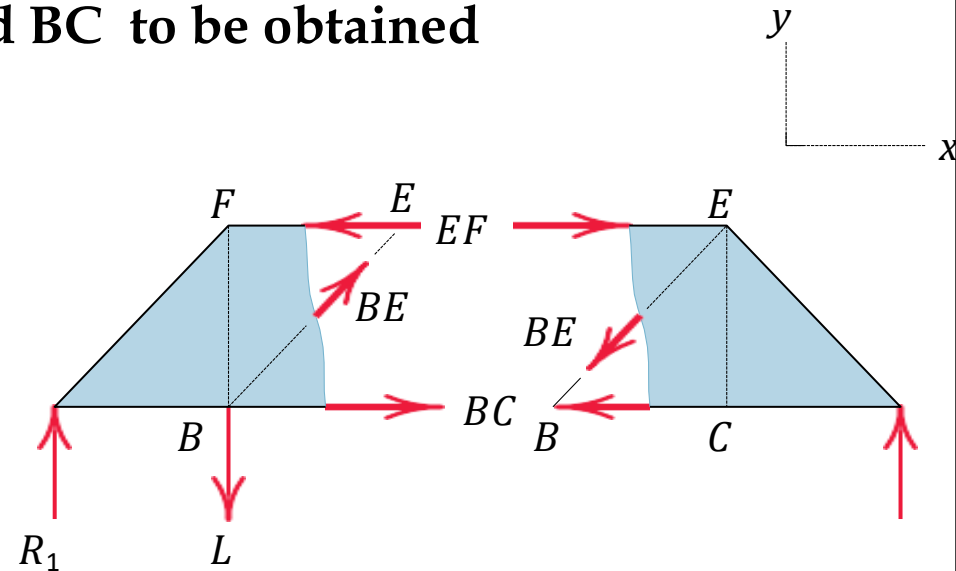
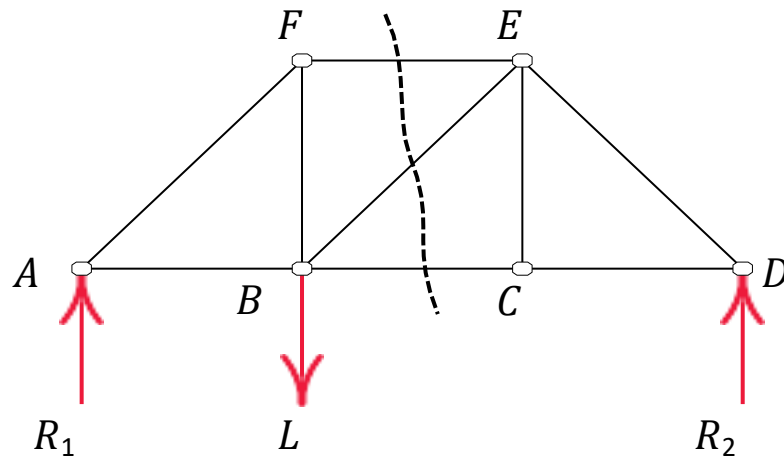
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Method of Section

- Use **Moment balance of forces** on section of Truss
- Entire section of the truss is considered a **single body in equilibrium**
- **Advantage:** Forces in **any desired member** to be found directly
- **Caution:** not **more than three members** whose forces are unknown should be cut

Method of section: Illustration

BE, EF and BC to be obtained



- Take Moment about
 - B to Find EF
 - E to Find BC
 - F to find BE

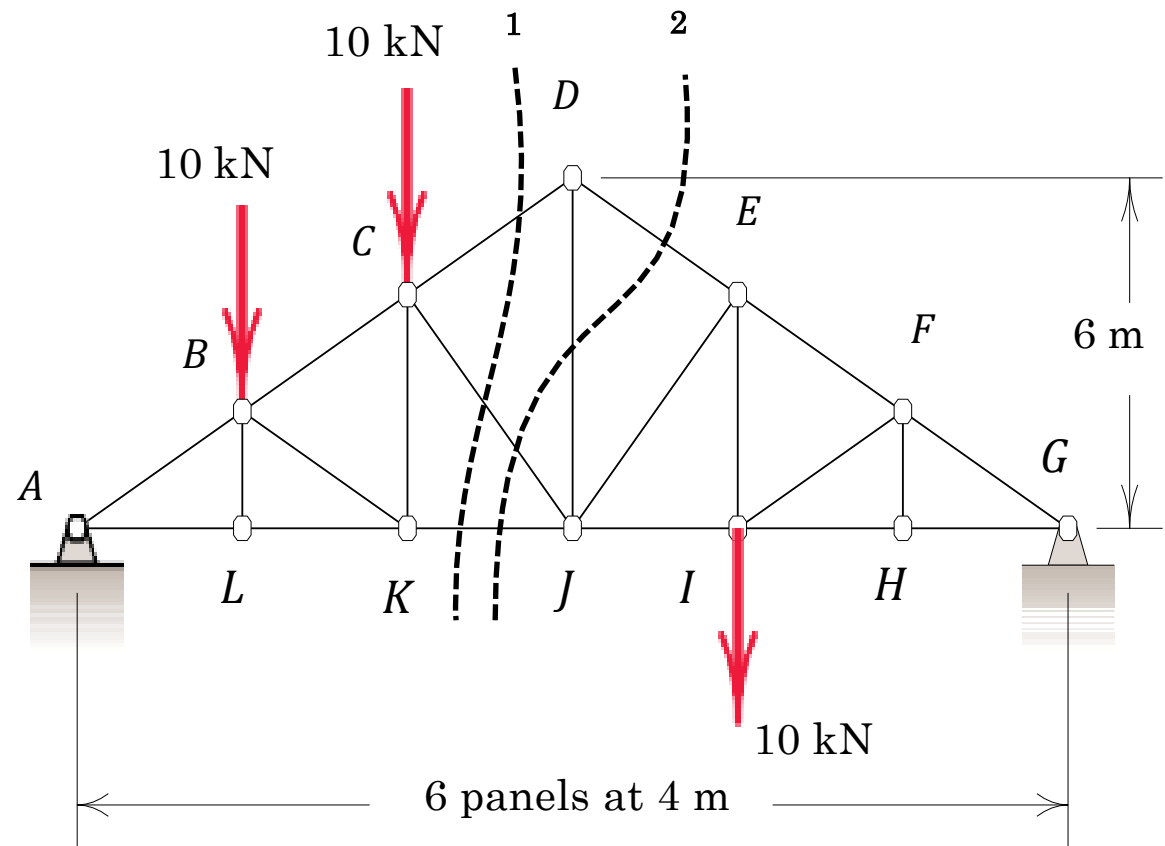
Additional Considerations

- **Moment equations** are used to great advantage
- Choose a **moment center**, either on or off the section, through which as many unknown forces as possible pass.
- A **positive answer** will verify the assumed sense, and a **negative result** will indicate that the force is in the sense opposite to that assume

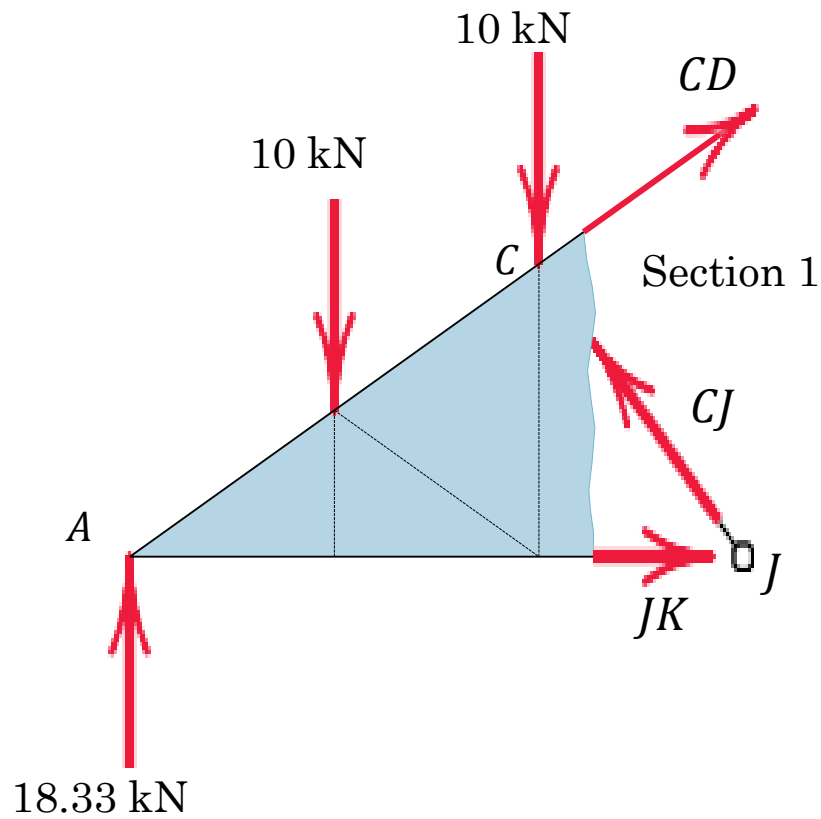
Additional Considerations

- Some cases the methods of sections and joints can be **combined** for an efficient solution.
- Section through the member can not be created without **passing through at least four unknown members**.
 - Determine the **forces in nearby members** by the method of sections and then
 - Progress to the **unknown member** by the method of **joints/section**.

Example 1: Calculate the force in member DJ of the Howe roof truss illustrated. Neglect any horizontal components of force at the supports.



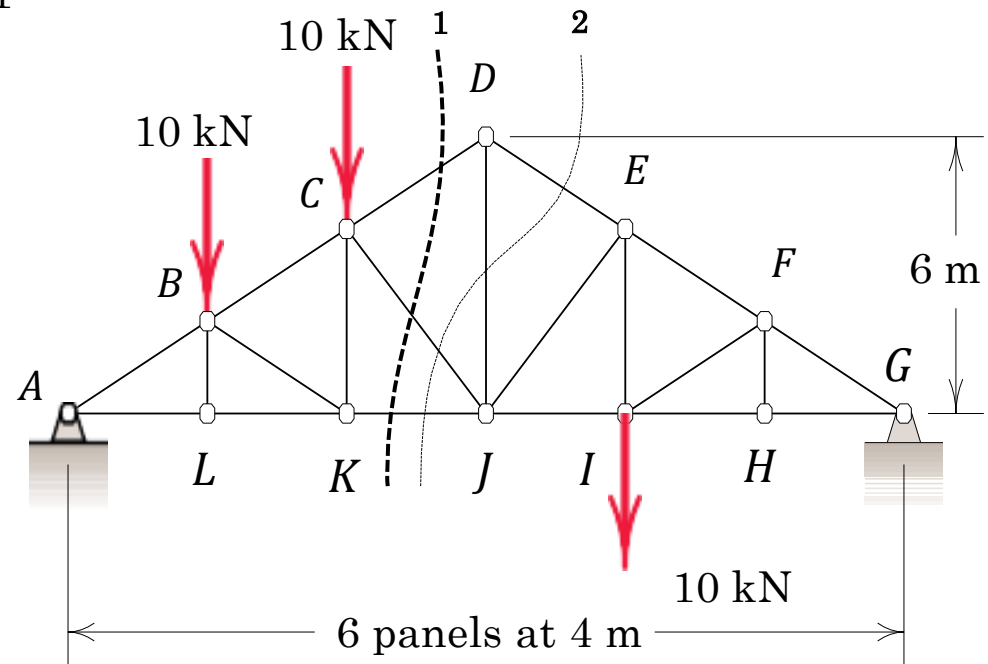
Example 1: Find CJ



$$[\Sigma M_A = 0]$$

$$0.707CJ(12) - 10(4) - 10(8) = 0$$

$$CJ = 14.14 \text{ kN C}$$



FRAMES AND MACHINES

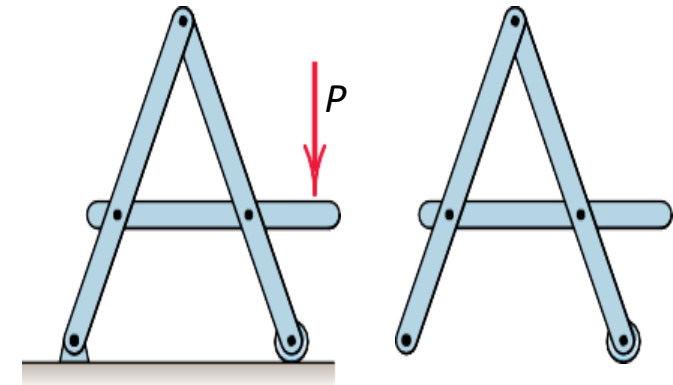
- A structure is called a **frame or machine** if at least one of its individual members is a **multiforce member, three or more forces**
- **Frames** are structures which are designed to **support applied loads** and are usually fixed in position.
- **Machines** are structures (static case) which contain **moving parts** and are designed to **transmit input forces or couples** to output forces or couples.

Interconnected Rigid Bodies

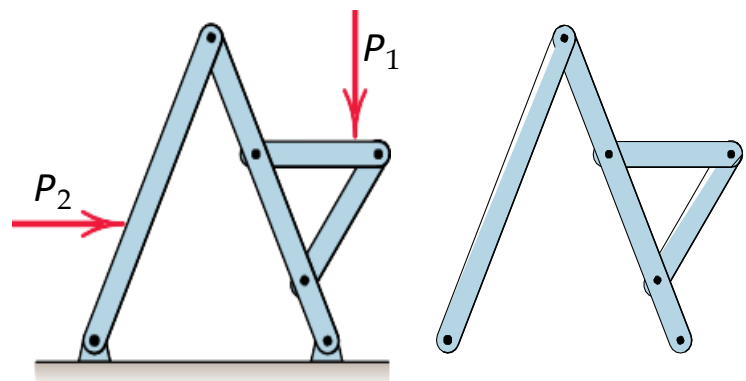
- Relies on **equilibrium of interconnected rigid bodies**
 - Members are **isolated** with a free-body diagram and
 - Apply the **equations of equilibrium** on isolated body
 - Principle of **action and reaction** must be carefully observed
- We may occasionally need to solve **two or more equations simultaneously** in order to separate the unknowns.

Collapsible vs Non collapsible frames

- **Non collapsible:** frame or machine constitutes a rigid unit when removed from its supports,
 - Analysis is begun by establishing all the forces external to the structure treated as a single rigid body
- **Collapsible:** frame or machine constitutes a non rigid unit when removed from its supports,
 - the calculation of the external support reactions cannot be completed until the structure is dismembered



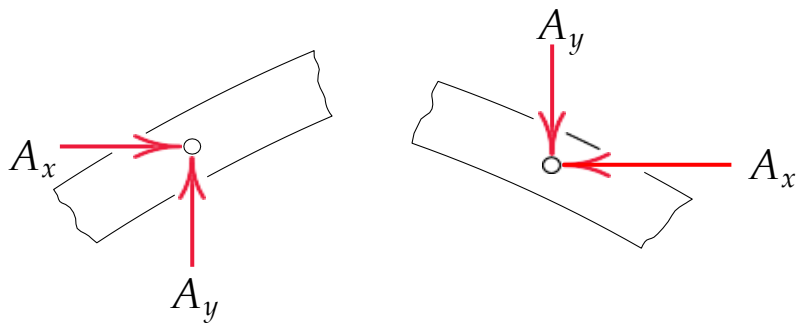
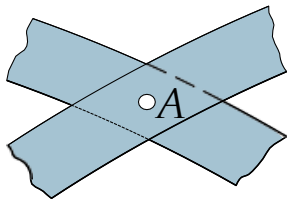
Rigid non collapsible



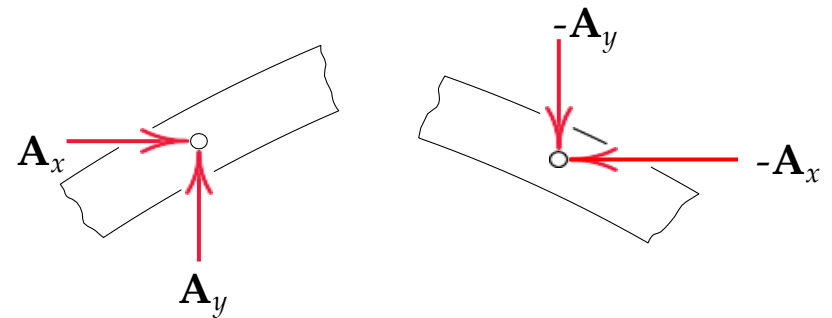
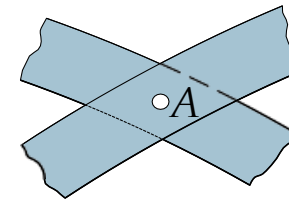
Nonrigid collapsible

Force Representation & Free-Body Diagrams

Scalar Notation

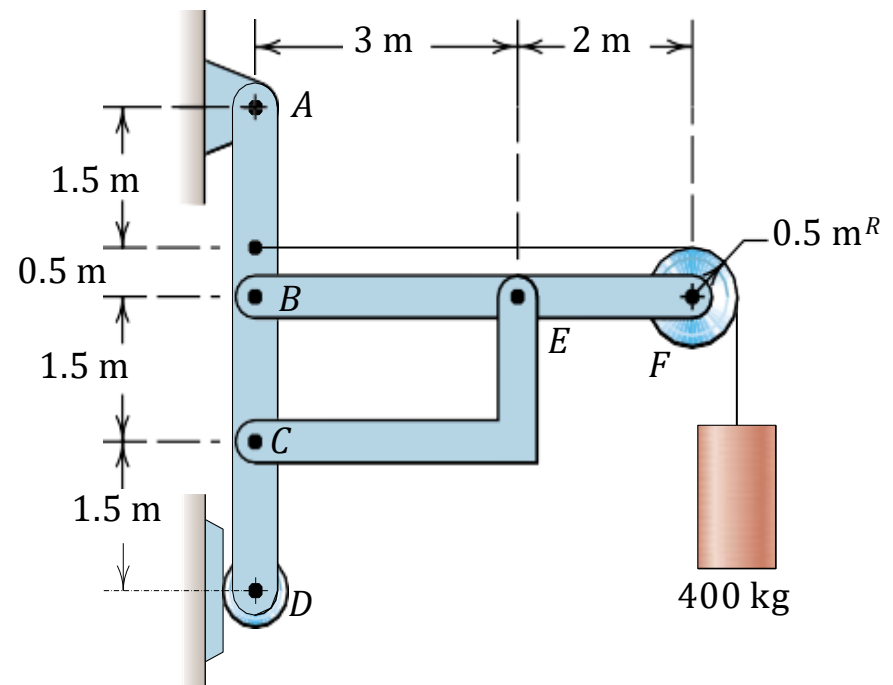


Vector Notation

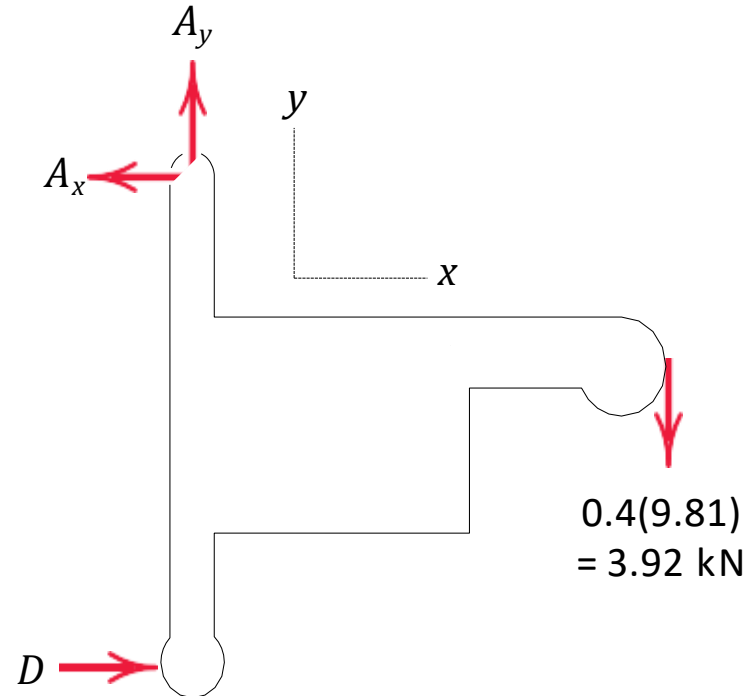
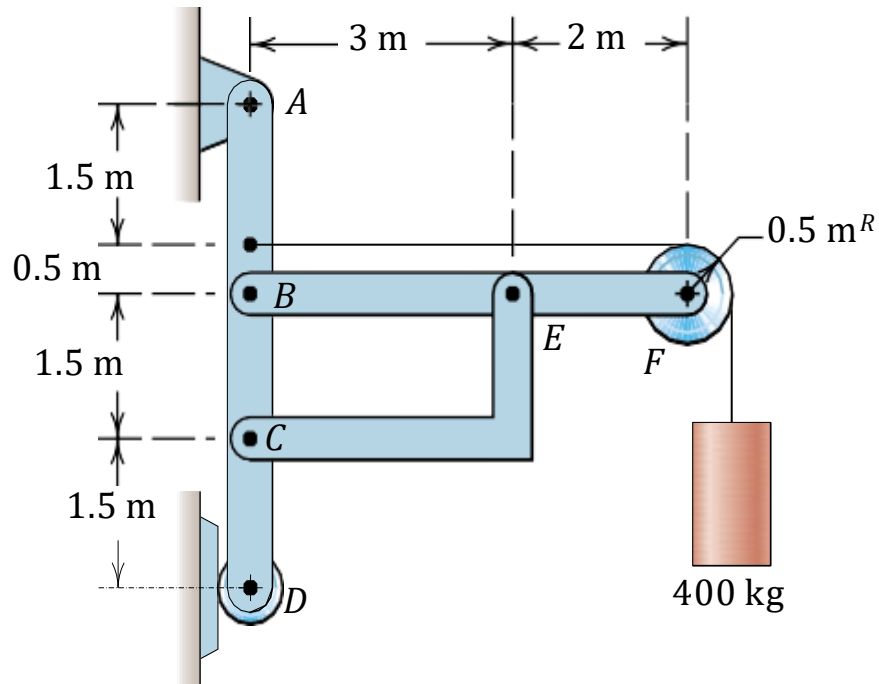


Use proper sign in vector notation

- Example 2: The frame supports the 400-kg load in the manner shown. Neglect the weights of the members compared with the forces induced by the load and compute the horizontal and vertical components of all forces acting on each of the members.



Example 2: Find Support Reactions



$$[\Sigma M_A = 0]$$

$$5.5(0.4)(9.81) - 5D = 0$$

$$D = 4.32 \text{ kN}$$

$$[\Sigma F_x = 0]$$

$$A_x - 4.32 = 0$$

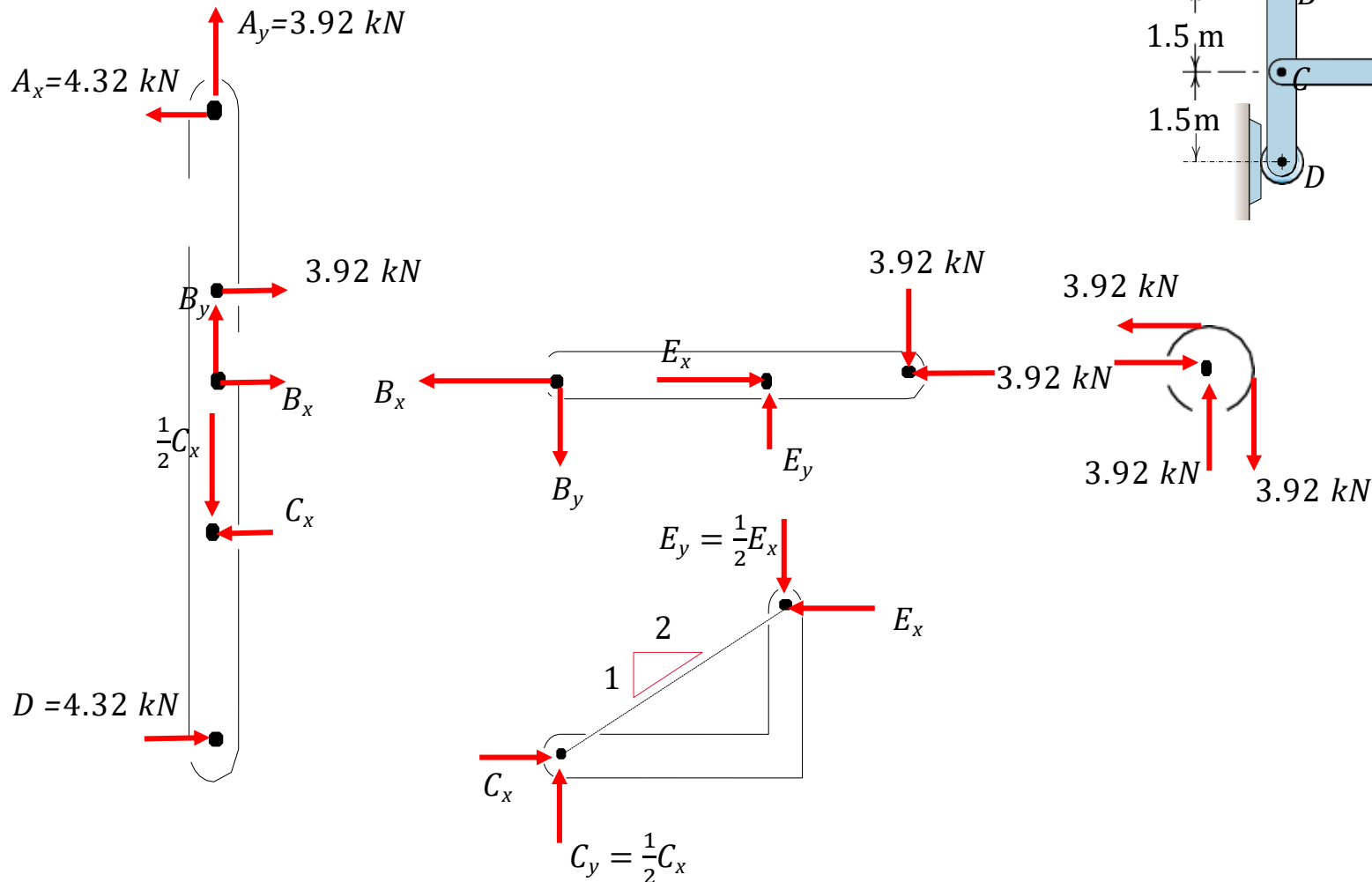
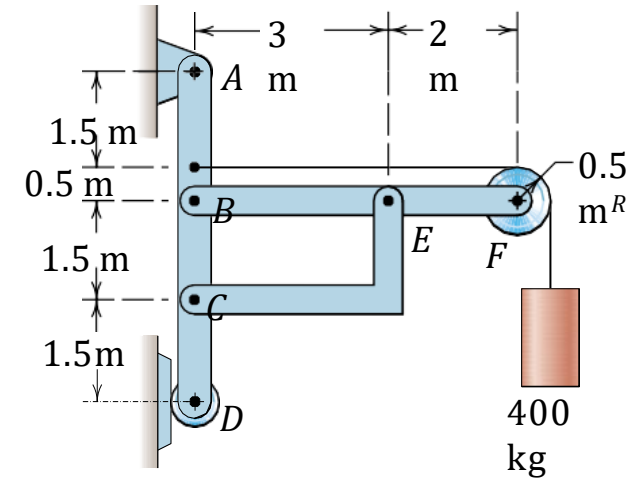
$$A_x = 4.32 \text{ kN}$$

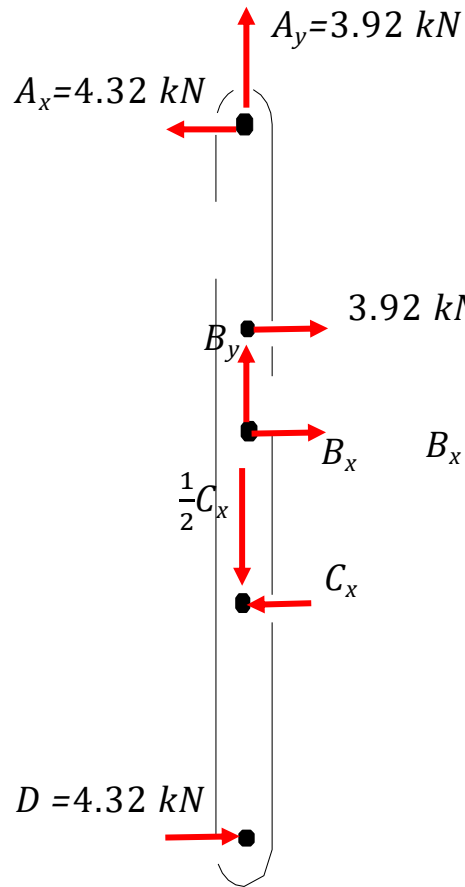
$$[\Sigma F_y = 0]$$

$$A_y - 3.92 = 0$$

$$A_y = 3.92 \text{ kN}$$

Find other reactions at joints





$$[\Sigma M_B = 0]$$

$$3.92(5) - \frac{1}{2}E_x(3) = 0$$

$$E_x = 13.08 \text{ kN}$$

$$[\Sigma F_y = 0]$$

$$B_y + 3.92 - 13.08/2 = 0$$

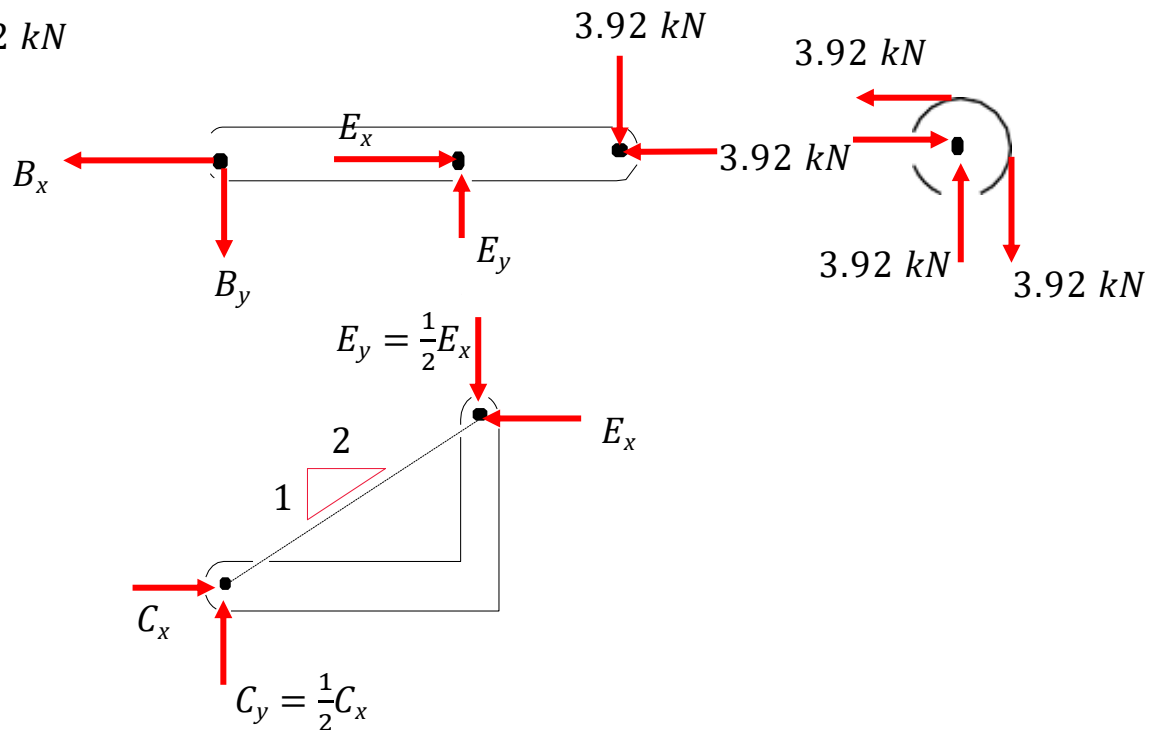
$$B_y = 2.62 \text{ kN}$$

$$[\Sigma F_x = 0]$$

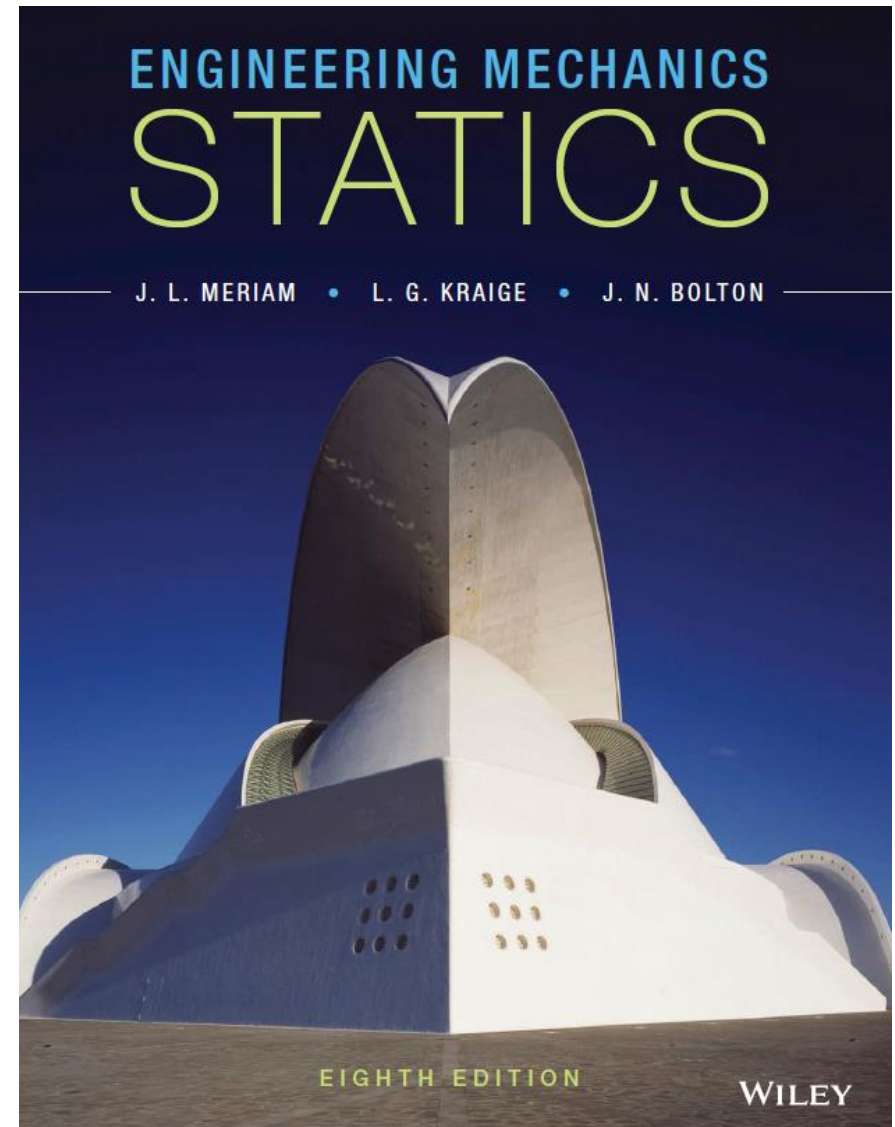
$$B_x + 3.92 - 13.08 = 0$$

$$B_x = 9.15 \text{ kN}$$

$$C_x = E_x$$



References :



THANK YOU